Activity Guide — middle / module-06-moon-phases-tithi

IKS Curriculum

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Activity 01 — Build a Tithi Calendar

Module: Moon Phases & Tithi · **Band:** Middle · **Day:** 4 · **Time:** 20 min (within Day 4's 45 min)

Type

Hands-on calendar build, paired

Goal

Each pair produces a tithi calendar for the current calendar month with every date labelled with its pakṣa, tithi, tithi end-time, and a small sketch of the expected Moon shape.

Materials per pair

- 1 printed page of the current month's *pañcāṅga* (teacher selects ONE source — Drik Panchang, Mypanchang, or any standard *pañcāṅga* — and uses it consistently across the class)
- 1 blank monthly calendar grid (printed handout — see below)
- 2 coloured pencils (yellow for *śukla*, blue/grey for *kṛṣṇa*)
- 1 black pen each

Set-up (teacher, before class)

- Print the *pañcāṅga* page for the current calendar month. One per pair.
- Print the blank calendar grid (template below). One per pair.
- Write the class's *pañcāṅga* source on the board so every pair uses the same one.

Instructions (student-facing — printable handout)

Build a Tithi Calendar — work in pairs.

1. Look at the *pañcāṅga* page for this month. Find today's date row.
2. Note the **tithi at sunrise** for today. Find the same tithi on your blank calendar grid for today.
3. Note the **tithi end-time**. After this time, the next tithi begins.

4. Repeat for every date in the month.
5. Colour-code: yellow for *śukla pakṣa* days, blue/grey for *kṛṣṇa pakṣa* days.
6. Circle any *amāvasyā* or *pūrṇimā*.
7. Sketch a small Moon shape next to each date — or every 3rd date if running short.
8. **At the bottom of the calendar, write:** the date of *amāvasyā*, the date of *pūrṇimā*, and any festival you can spot in this month.

Blank calendar grid (one per pair)

Month: ______________________________ Source: _______________________________

Date	Day	Pakṣa	Tithi	Ends at	Moon shape
1					
2					
3					
4					
5					
6					
7					
8					
9					
10					
11					
12					
13					
14					
15					
16					
17					
18					
19					
20					
21					
22					
23					
24					
25					
26					
27					
28					
29					
30					
31					

THIS MONTH:
- Amāvasyā falls on: _________________
- Pūrṇimā falls on: _________________
- Festival(s) I noticed: ____

Discussion prompts (last 3 min, whole class)

- Which date was the trickiest to figure out? Why?
- Did anyone find a tithi that "skipped a sunrise" (*kṣaya*)? Where?
- How does your calendar compare to the standard Gregorian calendar — what does it tell you that the Gregorian doesn't?

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Both partners reading the *pañcāṅga*, one calling out, one writing; colour-coding consistent	"OK, the 14th says caturdaśī, that's still kṛṣṇa pakṣa, so blue"
Needs intervention	One partner doing all the work; rows skipped; mixed-up colours	"I'll just write whatever" / "do we have to do every row?"

Safety / sensitivity

- No safety concerns specific to this activity.
- If a student is fasting on a tithi during class (some students observe *ekādaśī* fasts), be aware but don't single them out.

Differentiation

- **Tier up:** Compute one tithi from scratch using the elongation rule. Use an astronomy app to find the Sun's and Moon's ecliptic longitudes for a moment today. Subtract, mod 360, divide by 12, floor, +1 — that's the tithi number. Compare with what the *pañcāṅga* says.
- **Tier down:** Fill in the date column, day-of-week column, and tithi column only. Skip the end-times and Moon shapes for this lesson.

Clean-up

- Each pair keeps their completed calendar (paste into student workbook on Day 4 page).
- Teacher collects one duplicate copy for the project record.

Activity 1 — 10-Day Moon Observation Diary

Module: Moon Phases and Tithi · **Band:** Middle · **Duration:** 30 min

Purpose

Reinforce the day-by-day concepts of Moon Phases and Tithi through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

notebook, pencils, ~15 minutes per evening

What students do

Each evening for 10 days, students sketch the moon shape they see, note the time, the tithi name, and one feeling/observation. On Day 10 they compare sketches with the actual calendar.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (20 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.
- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Activity 02 — Diagram the Eclipse Geometry

Module: Moon Phases & Tithi · **Band:** Middle · **Day:** 7 · **Time:** 10 min (within Day 7's 45 min)

Type

Diagram drawing + short writing, paired

Goal

Each pair draws the Sun–Earth–Moon geometry for both eclipse types, marks the Moon's tilted orbit and its two nodes, and writes a 3-sentence explanation of why eclipses don't happen every month.

Materials per pair

- 1 printed handout (two partial diagrams — see below)
- 2 different colour pencils (one for the ecliptic plane, one for the Moon's tilted orbit)
- 1 black pen each

Set-up (teacher, before class)

- Print the handout. One per pair.
- Have a globe and a small dark ball ready for demo as students work.

Instructions (student-facing — printable handout)

Eclipse Geometry — work in pairs.

Below are two unfinished diagrams. Complete them.

Diagram 1: Solar Eclipse

The Sun is on the left. Earth is on the right. The Moon's orbit around Earth is drawn as a circle around Earth.

1. Place the Moon **between** the Sun and Earth. (Mark it with a small filled circle.)
2. Which tithi must this be? Label it: ________________.
3. Draw the Moon's shadow as two lines extending from the Moon to Earth. Shade the region of Earth that falls in the shadow.
4. Below the diagram, write: *"On this tithi, the Moon's shadow covers a small region of Earth. People in this region see a solar eclipse."*

[BLANK SPACE FOR DIAGRAM 1]

Diagram 2: Lunar Eclipse

The Sun is on the left. Earth is in the middle. The Moon's orbit is around Earth.

1. Place the Moon **opposite** the Sun, on the right of Earth. (Mark it with a small open circle.)
2. Which tithi must this be? Label it: ________________.
3. Draw Earth's shadow as two lines extending from Earth outward, away from the Sun. The Moon falls inside the shadow.
4. Below the diagram, write: *"On this tithi, the Moon passes through Earth's shadow. Anyone on the night side of Earth can see a lunar eclipse."*

[BLANK SPACE FOR DIAGRAM 2]

Bottom of the page

5. Now draw the Moon's orbit around Earth as a circle TILTED by 5° away from the Sun-Earth line. Mark the two **nodes** — the two points where the tilted orbit crosses the Sun-Earth line — with stars.

6. **In 3 sentences, answer:** Why don't eclipses happen every *amāvasyā* and every *pūrṇimā*?

Discussion prompts (whole class, 2 min after activity)

- Show your tilted-orbit diagram. Where on the orbit can eclipses happen? (Only at or near the nodes.)
- About how often does the Moon pass through a node? (Twice each month — but only twice each YEAR does the node alignment land near an *amāvasyā* or *pūrṇimā* — these are the “eclipse seasons.”)

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Both partners drawing carefully; shadow cones drawn at sensible angles; explanation in own words	“OK so if the orbit is tilted, the Moon is usually above or below the Earth’s shadow line”
Needs intervention	Both drawing the same circle on top of each other; explanation copied from textbook	Silence, or “I don’t get it”

Safety / sensitivity

- This is a paper activity. No safety concerns.
- Eclipse-viewing safety is taught in the lesson, not in this activity.

Differentiation

- **Tier up:** Add a third diagram showing the Moon at *amāvasyā* but FAR from a node — show how the Moon’s shadow misses Earth (passes above or below). This is the visual answer to “why not every month.”
- **Tier down:** Just complete Diagram 1 (solar) and Diagram 2 (lunar). Skip the tilted-orbit + nodes detail; the *concept* of “Moon’s orbit is tilted” is enough.

Clean-up

- Handouts paste into the student workbook (Day 7 page).
- Spare paper recycled.

Project Brief

Project Brief — Module 6 Moon Phases & Tithi (Middle)

Work sessions: Day 8 + Day 9 (in class) · **Presentation + summative:** Day 10 · **Total marks:** 40

The project

Each pair (or individual, with teacher approval) chooses **one** lunar phenomenon and produces a small project that shows:

1. **What the *pañcāṅga* / traditional Indian astronomy says** about this phenomenon. Must cite at least one source — the *Sūrya-siddhānta* (paraphrased), the class *pañcāṅga*, NCERT Physics XI, or NASA's published phase data.
2. **One modern astronomical account** of the same phenomenon — the underlying geometry, mechanism, or measurement.
3. **One personal observation or local example** — a date from your Moon diary, a festival from your family, a local *pañcāṅga* page from your area, etc.

Choose ONE lunar phenomenon

- **Moon phases** — the 8 named phases and the geometry that produces them.
- **Eclipses** — solar, lunar, why they happen at amāvasyā / pūrṇimā, and the 5° tilt.
- **Tithi calendar** — explain the tithi system, walk through a month's calendar.
- **Lunar months and festivals** — name the 12 months, link to festivals.
- ***Adhika māsa*** — the intercalary month and why a lunisolar calendar needs it.
- **Pañcāṅga** — the five limbs and what each tracks.
- **Cross-cultural lunar calendars** — Indian compared with Islamic / Chinese / Hebrew.

Why this matters

You're learning to hold two intellectual frameworks side-by-side — the *pañcāṅga* and modern astronomy. They are both real, both useful, and they answer slightly different questions about the same Moon. The discipline of treating both fairly is the deeper learning.

What success looks like

A strong project will: - Demonstrate the module's central concepts accurately (tithi, pakṣa, *amāvasyā*/*pūrṇimā*, the geometry). - Cite a real source — no inventions. - Connect to a modern astronomical account fairly — without claiming "ancient Indians knew this already." - Use a real observation from your own diary or calendar. - Be clearly communicated in your chosen medium.

Format options

Format	Best for topics	Notes
Poster (A2 or larger)	Any	Lots of visual space; works for nervous presenters
Live demonstration (2 min hands-on)	Phases (torch and ball), eclipses (shadow geometry)	Needs setup
Short written story (1 page)	Adhika māsa, cross-cultural	Read aloud or hand out
90-second video	Any	Pre-made and played in class
Slide deck (5–8 slides)	Pañcāṅga, calendars	Project from laptop
Infographic	Tithi calendar, month-festival mapping	Combine drawing + facts

Other formats with teacher approval. Whatever the format, the 3-minute time limit on Day 10 applies.

Working in pairs (or individually)

- **Pair work is encouraged**, especially for video/demo formats.
- **Individual is allowed** — fine for a single-topic explainer.
- Groups of 3 are acceptable but each member must have a distinct sub-role.

Schedule

Day	What you do	Submit at end of class
5	Choose topic + format + partner	Sticky note on the board
6	(Homework) Submit project plan paragraph	Project plan paragraph
8	Plan + start first artefact	Draft of first artefact (photo or scan accepted)
9	Refine + rehearse	Draft of final artefact
10	Present (3 min) + Q&A (1 min)	Final artefact + reflection slip

Citation rules

- At minimum **one** citation must appear in your project — visible on the poster, named in the video, or printed in the story.
- The citation must be from a real source: paraphrase of *Sūrya-siddhānta* (with chapter number — chapter 2 for tithi, chapters 4–5 for eclipses, chapter 14 for *adhika māsa*),

NCERT Physics XI, NASA's official lunar phase publications, the class *pañcāṅga* source.

- **Do not invent specific verse numbers.** If you don't have a verifiable verse, paraphrase and cite the text by name. Inventing a citation scores 0 on **Use of sources** and on **Originality**.

Rubric

See `rubric.md`. Read it before Day 8.

Self-reflection (submitted on Day 10 with project)

In 4–6 sentences, answer:

1. What did you set out to learn or make, and what did you actually end up with?
2. What is the most surprising thing you discovered while making this?
3. If you had three more days, what would you do differently?

Exemplars

A "Meets" exemplar will be added to `exports/pdf/middle-module-06-moon-phases-tithi/exemplar-strong.pdf` after the first cohort presents.